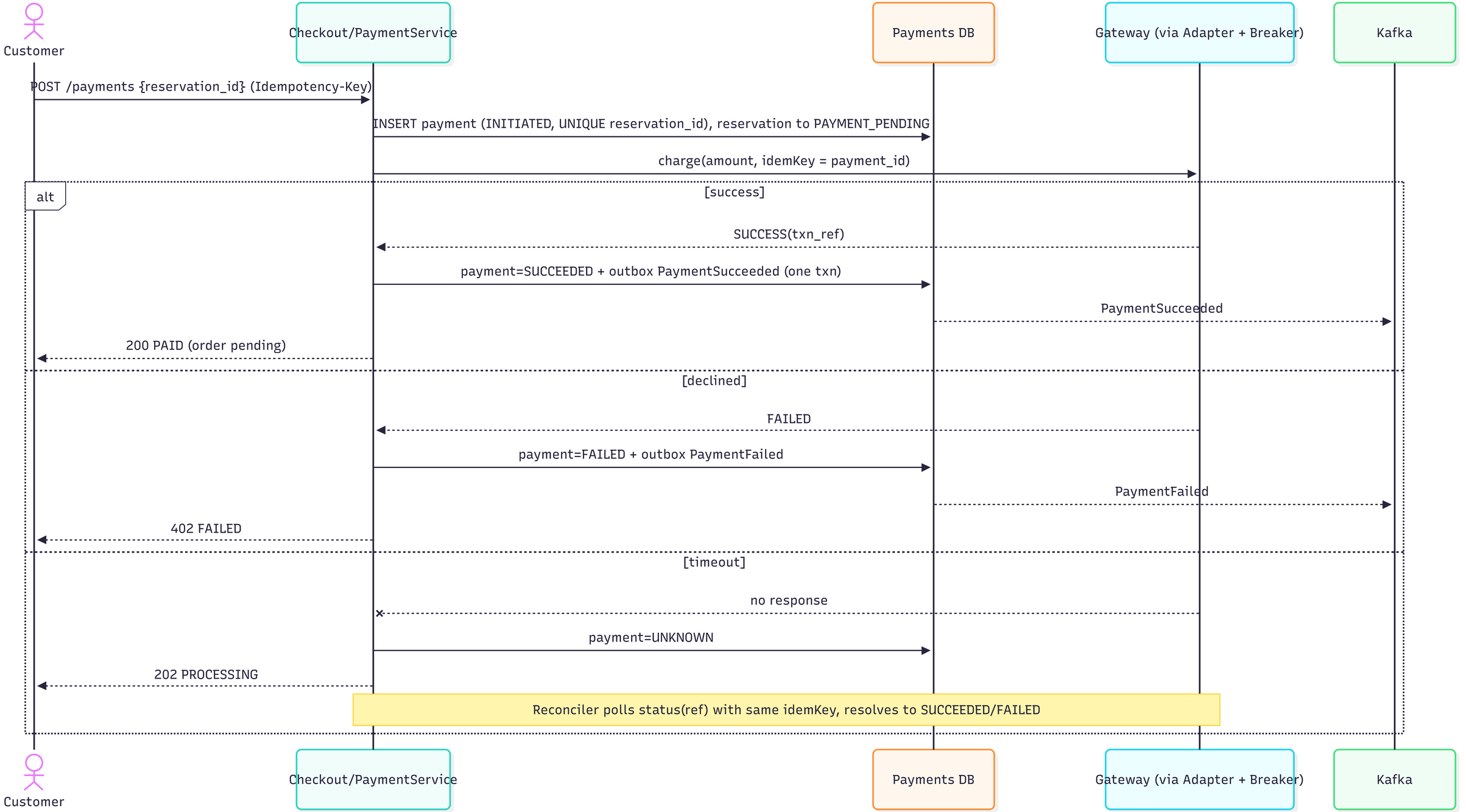

Customer
Checkout/PaymentService
Payments DB
Gateway (via Adapter + Breaker)
Kafka
POST /payments {reservation_id} (Idempotency-Key)
INSERT payment (INITIATED, UNIQUE reservation_id), reservation to PAYMENT_PENDING
charge(amount, idemKey = payment_id)
alt
[success]
SUCCESS(txn_ref)
payment=SUCCEEDED + outbox PaymentSucceeded (one txn)
PaymentSucceeded
200 PAID (order pending)
[declined]
FAILED
payment=FAILED + outbox PaymentFailed
PaymentFailed
402 FAILED
[timeout]
no response
payment=UNKNOWN
202 PROCESSING
Reconciler polls status(ref) with same idemKey, resolves to SUCCEEDED/FAILED
Customer
Checkout/PaymentService
Payments DB
Gateway (via Adapter + Breaker)
Kafka